Timothy N. Rubin, PhD

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Machine learning and data science leader with 10+ years of industry experience, including VP of Data Science at a public company. Led teams building business-critical ML models and infrastructure. Most recently founded and built a consumer AI application, leading the development, evaluation and optimization of production LLM systems. PhD with published research in probabilistic modeling and NLP (NeurIPS, JMLR).

EXPERIENCE

Founder & CEO · Wellness Technology Inc. 2024–present

Wellness AI: a mobile app for mental-health support combining AI chat and personalized meditation. Solo founded; own product, engineering, AI, analytics and growth, with part-time engineering and marketing contractors.

- Architected and built the production AI infrastructure: a backend running ~20 LLM pipelines (multi-phase chat with long-term memory, session insights, personalized guided meditations), alongside iOS and Android apps, with a privacy-first design that stores no user conversation data on the backend.
- Designed the prompts and evals behind every LLM-powered feature: custom evals per pipeline used to optimize prompts and models, plus production LLM-as-judge monitoring of every conversation for quality, safety and engagement, with human review of flagged sessions.
- Built the analytics and experimentation stack: A/B rollouts of model and prompt changes, dashboards tracking engagement, retention, subscriptions and judged interaction quality, plus validation of quality and engagement metrics against user ratings and mood scores.
- Grew the app to 250+ paying subscribers, quadrupling subscribers and tripling revenue from March to September 2026. Doubled LTV per install through pricing, paywall and acquisition optimization; achieved 4.9★ ratings on both app stores and top-decile conversion against industry benchmarks.

AI Consultant · Arvo Labs & Independent 2024–2025

- Part-time AI/ML advisor to an agency founded by former Doma colleagues: feasibility reviews, LLM evaluation metrics, model prototyping.

Sr. Manager → Director → VP of Data Science · Doma 2020–2023

- Led a DS/ML team of up to seven data scientists, ML engineers and analysts, responsible for technical direction, hiring and people development; partnered directly with executives on ML and product strategy.
- Owned the instant-title risk model that supported ~80% of $20–30M in annual enterprise revenue, with loss ratios significantly below targets; rebuilt its training, evaluation and loss-projection infrastructure.
- Led the technical case and review process with Fannie Mae that secured approval for Doma's Title Waiver program, covering production performance, stress-tested loss estimates and model fairness.
- Grew the team from 2 to 7 across data science, ML engineering and analytics; drove five promotions in three years, including two into management.

Senior Data Scientist / MLE → Data Science Team Lead · Change Healthcare 2018–2020

- Built NLP models that classified claims from long clinical PDFs to reduce manual review; deployed in the production claims pipeline.
- Built and led development of a reusable data-labeling platform and model training and deployment pipeline, adopted by other teams across the ~30-person ML group.
- As team lead, set technical direction and priorities for ML engineers and data scientists working with clinical text and claims data.

Senior Data Scientist · SurveyMonkey 2016–2018

- Built the algorithmic core of SurveyMonkey Genius, the company's first customer-facing ML product, predicting survey completion rates and times, trained on millions of surveys and tens of millions of responses, plus personalized recommendations.
- Created an internal use-case ontology and automated classification model, driving sales assistance and user-personalization opportunities.

Senior Research Scientist · Indiana University 2012–2016

- Led and collaborated on ML research published in NeurIPS, JMLR and Machine Learning, including a novel topic model for jointly analyzing text and spatial data from neuroscience publications, empirical evaluations of semantic models, and improved Bayesian estimators for LDA.

Graduate Researcher · University of California, Irvine 2006–2012

- Developed and published probabilistic ML models for multi-label document classification, recommendation systems, and concept learning.

EDUCATION

PhD & MA, Cognitive Sciences · University of California, Irvine 2012, 2009

BS, Cognitive Science · Tufts University 2004

EXPERTISE

Machine learning and data science: Machine learning and NLP (predictive modeling, unsupervised learning, Bayesian methods), LLM product development and evaluation (LLM-as-judge design and calibration, human evaluation), experimental design, statistics, causal inference, KPI design and dashboarding

Product and leadership: Team hiring and development, executive and partner communication, product ownership from roadmap to pricing and growth, contractor management

Engineering and tools: Python (scientific stack), SQL, Spark, R, FastAPI, Azure, Firebase, Looker; agentic development with Claude Code and Codex

SELECTED PUBLICATIONS & PATENT

13 papers, ~1,000 citations in venues including NeurIPS and JMLR.

- Rubin, Koyejo, Jones & Yarkoni (2016). Generalized correspondence-LDA models for identifying functional regions in the brain. *NeurIPS*.
- Papanikolaou, Foulds, Rubin & Tsoumakas (2017). Dense distributions from sparse samples: improved Gibbs sampling parameter estimators for LDA. *JMLR*.
- Rubin, Chambers, Smyth & Steyvers (2012). Statistical topic models for multi-label document classification. *Machine Learning*.
- Mason & Rubin (2022). Predictive time series data object machine learning system. US Patent 11,580,309.